

Mohamed Zeina

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EXPERIENCE

Embedded Systems & Full Stack Software Engineer (Contract)

August 2025 – Apr. 2026

Energco, ZAIM Teknopark

Istanbul, TR

- Built a multi-drone system coordinating 5 UAVs with real-time 30 FPS video streaming, live telemetry via WebSockets, and fault-tolerant reconnection and failsafe handling
- Developed a Ground Control Station using React and Python FastAPI with real-time map interface, manual flight controls, and remote GPS navigation
- Implemented YOLOv8 object detection with PID-controlled flight for real-time subject tracking; calculated target GPS from gimbal angles, GPS, and rangefinder data to autonomously dispatch mini drones
- Configured bridged wireless network between GCS and drones; set up embedded Linux on Raspberry Pi for video streaming and MAVLink communication

Teaching Assistant for Introduction to Computing

Sep. 2021 – Jan. 2024

Sabancı University

Istanbul, TR

- Tutored over 50 students weekly in C++ programming; designed and evaluated 30+ assignments and exams aligned with course learning objectives

Full Stack Software Engineer Intern

July 2019 – Sep. 2019

Interact Technologies

Istanbul, TR

- Implemented TCP/IP protocol achieving 99.9% data transmission success and reduced GUI load time by 40% through code optimization
- Designed a NoSQL document database schema supporting 500+ patient records, enabling downstream reporting

EDUCATION

Sabancı University

Istanbul, TR

Master of Science in Computer Science and Engineering (M.Sc.CSE)

Sep. 2021 – Jan. 2024

- GPA: 3.91/4.0 | Full Scholarship | Thesis: Built a framework combining GANs, SMOTE, and differential privacy to preserve household privacy in smart meter data

Sabancı University

Istanbul, TR

Bachelor of Computer Science and Engineering (B.Sc.CSE)

Sep. 2016 – June 2020

- GPA: 3.73/4.0 | Half Tuition Scholarship | Degree of High Honor (2016–2017, 2018–2020)

PROJECTS

[Noblecart](#) | [GitHub](#) | *Node.js, Express.js, MongoDB, EJS, Three.js, Stripe, Cloudinary, PDFKit*

- Built a full-stack MVC e-commerce app with session-based auth, bcrypt password hashing, CSRF protection, Helmet security headers, and role-scoped admin controls with Cloudinary image and GLB 3D model storage
- Implemented an interactive Three.js 3D product viewer with OrbitControls, a slide-out cart drawer with live quantity controls and badge updates via Fetch API, Stripe Checkout for payments, and server-side PDF invoice generation with PDFKit

[Hearsay](#) | [GitHub](#) | *Next.js, TypeScript, PostgreSQL, NextAuth, NextUI, Tailwind, Prisma, Zod*

- Built a full-stack threaded discussion app with GitHub OAuth, topic-scoped posts, recursively nested comments with reply-aware delete logic, live typeahead search across topics and posts, and authorization-checked Zod-validated mutations
- Architected on Next.js App Router with React Server Components and Suspense streaming for parallel post/comment/sidebar loading, React `cache()` for per-render query deduplication, and a custom Tailwind design system with deterministic slug-hashed topic theming and shared primitives (`PostCard`, `SurfacePanel`, `ThreadMap`)

[Node Social](#) | [GitHub](#) | *Node.js, Express.js, MongoDB, GraphQL, JWT, Cloudinary, React*

- Built a GraphQL API with JWT authentication, server-side input validation, paginated post feeds, and image upload via Multer and Cloudinary
- Implemented ownership-enforced CRUD mutations, time-limited tokens with automatic client-side expiry, and real-time communication with Socket.io

[JSNote](#) | [GitHub](#) | *React, TypeScript, Vite, esbuild-wasm, Monaco, Redux, Immer, Express, npm workspaces*

- Published `jsnote-zeina` to npm as a CLI-distributed in-browser JS/JSX notebook with Monaco IntelliSense via `@typescript/ata`, esbuild-wasm bundling with unpkg npm resolution, sandboxed iframe execution, and cumulative cell scope sharing
- Architected as a three-package npm workspace with Redux/Immer state, debounced persistence, multi-notebook directory mode, find/replace, pinned cells, runtime-error cell attribution via source-map-js, and Vitest + Playwright test coverage

SKILLS

Languages: C++, C#, Python, Java, JavaScript, TypeScript, SQL

Frameworks & Libraries: React, Redux, Next.js, Node.js, Express, Vite, Mongoose, GraphQL, Prisma, Socket.io

Web & Databases: Tailwind CSS, RESTful APIs, WebSockets, MongoDB, PostgreSQL, SQLite, Redis

Testing: Vitest, Playwright, Jest

AI Tooling: Claude Code, Multi-agent Orchestration, Prompt Engineering, LLM Integration

DevOps: Docker, Nginx, Kubernetes, AWS (EC2, RDS, S3), Google Cloud (GKE), CI/CD (GitHub Actions, Travis CI)

Embedded & CV: Raspberry Pi, MAVLink, UDP Streaming, OpenCV, YOLOv8

Spoken: Arabic (Native), English (Fluent)